Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/15/2016
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Fundamental Research on International Standard of Fire Safety in Space - Subteam 2: Wire Combustion with External Radiation in Support of the JAXA Project Fundamental Research on International Standard of Fire Safety in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Japanese Space Agency (JAXA) AO for Fundamental Research on an International Standard of Fire Safety in Space
Start Date:	03/06/2014	**End Date:**	03/05/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Prof. Carlos Fernandez-Pello is U.S. Co-Investigator on Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA PI is Prof. Osamu Fujita, Hokkaido University.		
COI Name (Institution):			
Grant/Contract No.:	NNX14AF01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Prof. Fernandez-Pello's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012. The objective of the proposed research program is to continue the experimental study of the flammability of wire materials in space exploration atmospheres and associated computational/theoretical tools to aid interpretation of test results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in space allows us to accurately control the flow field and thus elucidate the importance of a critical Damkohler number (flow time /reaction time) on flame extinction. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Investigation and results have Earth benefits for terrestrial fire safety.
Task Progress:	During this reporting period the research extended the previous work to determine the normal gravity and microgravity flammability limits of ETFE (ethylene tetrafluoroethylene) insulated wires to the study of the flammability limits of PE (polyethylene) insulated wires exposed to external radiation under normal gravity. PE has very different physicochemical properties (e.g., fire resistance and melting propensity) from ETFE; thus, it is a good material to compare the effect of insulation composition on the wire flammability. Experiments were conducted to find the flammability characteristics for two sizes of wires: (1) 9-mm wire diameter with 5.5 mm copper core and (2) 8-mm wire diameter with 3.5 mm copper core. Both LDPE (low-density polyethylene) and HDPE (high-density polyethylene) insulation are studied. Both insulations are visually transparent, while the HDPE has a lower degree of branching in micro-structure, slightly higher density, and higher stiffness than those of LDPE. The experiments consisted on the determination of the ignition and flame spread limiting boundaries of the wires in oxygen concentrations ranging from 10 to 21% (air) and external radiant fluxes from 0 to 34 kW/m2. The apparatus and supporting equipment is designed following similar concepts as those for the MIST microgravity experiments to be conducted in the International Space Station. The apparatus consists of a flow duct, infrared heaters, and supporting instrumentation. The test section itself consists of a 254 mm long Pyrex cylinder with an inner diameter of 70 mm. The PE insulated wire is placed vertically at the centerline of the cylinder, supported at its bottom by an aluminum cylinder of the same diameter. Below the test section, a flow chamber allows for controlled flows of modified air to pass through borosilicate beads, homogenizing the flow and evenly mixing the gases. The air flow uses a prescribed oxygen (balance nitrogen) concentration of between 12 and 21% at a fixed velocity of 4 cm/s at 1 atm. Equispaced around the Pyrex cylinder at 120 degree intervals are three IR lamps. These radiant heaters are quartz infrared halogen lamps with a lit length of 127 mm and a total length of 220 mm, fitting into parabolic strip heaters. Irradiation to the fuel sample is calibrated to be approximately uniform with an overall incident flux up to 34 kW/m2. For ignition, each wire sample is first preheated under the external radiation and opposed flow for 1 min. Then, a small premixed methane flame is placed on the top of wire for another 2 min to ensure a same ignition for all experiments. Experiments consisted in determining whether opposed flame spread occurred along the wire insulation for 100 mm under a given set of ambient oxygen and external radiation. Increasing the oxygen concentration and external radiation results in a change from “no spread” to “spread,” which is defined as the flammability limit. Moreover, in order to study the influence of copper core, additional experiments using a thin hollow stainless steel as core material were conducted. The thin hollow stainless steel has very low thermal conductivity and while behaving similar to air can prevent the bending of soft LDPE once heated. In the absence of an external radiant flux, the Limiting Oxygen Concentration (LOC) for HDPE and LDPE insulated wires were found to be the same but dependent on the core diameter: 21% for the 9-mm diameter wire (5.5 mm core) and 19% for the 8-mm diameter wire (3.5 mm core). Additional experiments with the same 9-mm and 8-mm wires with core of thin hollow stainless steel showed a lower LOC of 17.5%. These findings suggest that the copper core acts as a heat sink to reduce the wire flammability. Compared to the LOC of 31.5% found for ETFE insulated wires, PE insulated wires have a substantially higher fire risk. As the external radiation increases, the oxygen concentration limit for all tested wire configurations decrease in a linear manner, defining the flammability boundaries. A phenomenological explanation of the results is that the cooling from the copper core reduces the effective heating from the flame and the external radiation, making it necessary to increase the flaming heating via increasing the environmental oxygen concentration and consequently the flame temperature. In addition, the flammability boundary was found to be approximately linear and have the same slope (radiant flux/oxygen concentration) for both 9-mm and 8-mm diameter wires with LDPE insulation. Comparison also showed that the absolute slope was smaller for HDPE insulted wires while was larger for wires without copper core. Moreover, experiments showed that there was a minimum oxygen concentration for wires, below which flame cannot be sustained even under intensive radiation. The value of minimum oxygen concentration was 14.5% for both 9-mm and 8-mm wires with the copper core and either LDPE or HDPE insulation. Such minimum value decreased to 12.5% if the copper core was removed. This result further confirmed the role of copper core as a heating sink. Compared to previously found value of 27% for a thinner 1.1-mm ETFE insulated wire, these PE insulated wires have a substantially lower minimum oxygen concentration despite of larger cores. Therefore, it appears that the degradation chemistry of polymer insulation plays an important role in wire flammability. Furthermore, different melting phenomena were observed in experiments during the ignition and spread of flame over both LDPE and HDPE insulated wires. Compared to non-melting polymers like PMMA and ETFE, the dripping of molten polymers like PE bring new fire safety issues. Experimental observations showed that the degree of melting and frequency of dripping were greater for LDPE insulation than those for HDPE insulation. Such difference in melting is not only responsible for the difference in flammability limits, but also results in different overall fire risks for wire. This suggests that the small difference in insulation micro-structure like in the case of LDPE and HDPE should not be neglected.

Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	Miyamoto K, Huang X, Hasimoto N, Fernandez-Pello C, Fujita O. "Flammability limit of polyethylene insulated wires under varying oxygen concentrations and external radiation." 36th International Symposium on Combustion, Seoul, Korea, July 31-August 5, 2016. Submitted as of December 2015 to the 36th International Symposium on Combustion, Seoul, Korea, July 31-August 5, 2016. , Dec-2015